Movable Object Detection in Remote Sensing Images via Dynamic Automatic Learning

Xiang Zhang, Hangzai Luo, Sheng Zhong, Lei Tang, Jinye Peng, and Jianping Fan

***Abstract*— The performance of deep networks for object detection in remote sensing images (RSIs) largely depends on the availability of large-scale training images whose labels are given at the bounding-box level through a labor-intensive manual labeling process. To alleviate the huge burden of providing bounding-box annotations manually for movable objects, we propose a new approach, called dynamic automatic learning (DAL), to progressively learn object detectors. Specifically, a novel initial annotation generation (IAG) strategy is first designed to produce bounding-box annotations for movable objects in multitemporal RSIs. During this process, image-level labels need to be manually labeled for the generated candidates. Next, a detection network learns the detection knowledge from multitemporal RSIs with bounding-box annotations and then transfers the knowledge to generate pseudoboxes for the unlabeled data. Finally, with these pseudoboxes, the object detector can be optimized for generating accurate pseudoboxes iteratively. Furthermore, we introduce a pseudobox filtering (PBF) strategy to purify the quality of pseudoboxes to obtain accurate supervision. Our experiments on the challenging Northwestern Polytechnical University (NWPU) VHR-10.v2 and object detection in optical remote sensing images (DIOR) datasets have demonstrated that our DAL approach can achieve competitive results compared with state-of-the-art methods.**

***Index Terms*— Automatic learning, deep networks, object detection, remote sensing images (RSIs).**

I. Introduction

DEEP convolutional neural networks (DCNNs) have demonstrated impressive performance in remote sensing object detection (RSOD) [1], [2], [3], [4], [5], [6], [7], [8]. The success of such object detectors largely relies on a large amount of training data which requires manually annotated bounding boxes. However, manually labeling large-scale precise bounding-box annotations is time consuming and expensive. Moreover, labeling small and dense objects in remote sensing images (RSIs) is prone to errors and difficult to ensure consistent labeling quality.

To reduce the labeling cost, numerous methods [9], [10], [11], [12], [13], [14], [15], [16] are devoted to studying image-level weakly supervised object detection (WSOD) techniques, where the training images need only image-level labels. The common pipeline of most WSOD methods is to first produce initial training samples and then use them to train the object detector and augment training data iteratively until the model converges.

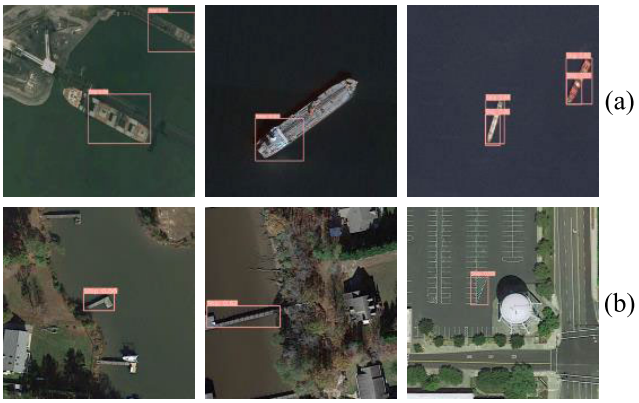


Fig. 1. Unreliable bounding-box annotations of training instances. (a) One part of the object. (b) Confusing backgrounds.

Due to the supervision gap between fully (i.e., bounding-box annotations) and weakly (i.e., image-level labels) supervised object detection, WSOD faces the following key challenges: 1) typical feature representations learned from the weakly supervised setting are only representative of some salient parts of objects, which deviates from a desirable detection result; 2) the model is difficult to converge when unreliable labels (see Fig. 1) of training instances are introduced in the iterative learning process.

To overcome the above-mentioned limitations, we propose a novel approach, named dynamic automatic learning (DAL), to progressively learn object detectors. The proposed DAL is essentially a simple-to-complex object detection framework. An initial detector can be learned from multitemporal RSIs with simple backgrounds. Furthermore, based on the initial detector, more objects from complex images can be detected so that a more powerful detector can be continually learned for object detection. It performs pseudolabeling for unlabeled images and trains a detector using these pseudolabels along with some initial training samples with bounding-box annotations at each iteration. Specifically, bounding-box annotations of initial training samples are generated first by feeding multitemporal RSIs containing movable objects into our proposed initial annotation generation (IAG) strategy. The initial training samples with bounding-box annotations are then used to learn detection knowledge by an object detector. Next, the learned detector transfers the detection knowledge to produce pseudoboxes for the unlabeled data. Finally, with these pseu-

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doboxes, the detector can be optimized to generate accurate pseudoboxes iteratively. Moreover, an effective pseudobox filtering (PBF) strategy is introduced to purify the quality of pseudoboxes to obtain accurate supervision. The main advantage of our proposed method is that it introduces the IAG strategy to solve the lack of instance-level bounding-box supervision problem and adopts the PBF strategy to alleviate the drift problem caused by noisy labels. It should be emphasized that our proposed method does not require any labels of the target data, but requires manual labeling of object categories for the generated candidates during the process of generating bounding box-level annotations for initial training samples.

Our main contributions are summarized in the following.

1) We propose a novel IAG strategy to generate high-quality bounding-box annotations for initial training samples. Furthermore, a PBF strategy is designed to purify the quality of pseudoboxes, thereby improving network performance.
2) We develop a DAL approach that progressively incorporates the results of the trained detector to improve the performance of object detection.
3) Our experiments on the challenging Northwestern Polytechnical University (NWPU) VHR-10.v2 [17] and object detection in optical remote sensing images (DIOR) [4] datasets have demonstrated that our proposed DAL can achieve outstanding performance compared with existing WSOD techniques. Besides, our DAL can achieve better performance in some classes (e.g., airplanes) than most existing fully supervised approaches.

II. Related Work

With the boom of DCNNs, object detection has been extensively studied for RSIs. Tang et al. [18] improved the faster R-CNN architecture by applying a hyper region proposal network and a cascade of boosted classifiers. Tian et al. [19] introduced an embedded graph attention network (EGAT) to exploit the underlying spatial and semantic relations among objects to boost the performance of RSOD. Chalavadi et al. [20] designed a new network called mSODANet, which improves the performance of object detection in aerial images by exploiting hierarchical dilated convolutions. Other works [21], [22], [23], [24] are also devoted to exploring object detection in the field of RSI understanding. Recently, oriented object detection has also been widely studied in RSIs. Successful examples include oriented R-CNN [25], lightweight oriented object detector (LO-Det) [26], General Gaussian Heatmap Label Assignment (GGHL) [27], Anchor-free Oriented Proposal Generator (AOPG) [28], and RSDet++ [8]. In particular, Huang et al. [26] proposed a LO-Det, which uses channel separation–aggregation structures to simplify the deep model and can produce competitive results. GGHL [27] is an arbitrary-oriented object detection method. It can refine the features of the rotating object by introducing the 2-D Gaussian label distribution based on fully convolutional one-stage object detector (FCOS) [29]. However, these fully supervised object detection approaches are data-driven and rely on accurate and reliable bounding-box annotations.

Recently, the semisupervised learning technique is developed to alleviate the heavy labeling burden. Semisupervised learning approaches improve the performance of object detection by leveraging a massive amount of unlabeled data to augment the limited labeled data. For example, Chen et al. [30] designed a generative adversarial network to extract data distribution. The extracted information is then used to enhance feature representation. Uehara et al. [31] introduced a human cooperative semiself-training (SST) framework, which combined self-training and user intervention to effectively detect objects in satellite images. Zhang et al. [32] proposed an improved self-training combined with object first mixup (OF-mixup) to improve the semisupervised detection performance of RSIs. Although these approaches have achieved reasonable performance, they still cannot escape the limitation of manual labeling [33].

To further reduce the human involvement in labeling bounding-box annotations, some weakly supervised detectors are trained by only using image-level labels. In particular, Zhou et al. [34] proposed a new WSOD framework to improve detection accuracy by integrating negative bootstrapping and transferred deep features. Zhang et al. [35] proposed a CNN-based method that can automatically mine and augment training images to improve aircraft detection accuracy. Yao et al. [12] proposed a curriculum learning strategy to progressively learn detection knowledge. Feng et al. [11] constructed a novel architecture named TCANet that improves the performance of WSOD by leveraging GCAE and DLCR modules to learn complementary and discriminative visual patterns. Cheng et al. [36] designed a self-guided proposal generation (SPG) approach, which can generate high-quality proposals to enhance the performance of WSOD in RSIs. Qian et al. [15] proposed a novel metric called objectness score (OS), which can be integrated into the training loss of the WSOD model to improve the performance of RSOD. Although these WSOD methods have made great progress, they fail to achieve a satisfying result compared with fully supervised methods due to the lack of instance-level bounding-box supervision and suffering from model drift problems.

Different from the previous works, we design an effective strategy to generate high-quality bounding-box annotations for initial training samples (i.e., multitemporal RSIs containing movable objects) to solve the lack of instance-level bounding-box supervision problem and introduce a simple PBF strategy to remove noisy pseudoboxes to alleviate the model drift problem during iterative training.

III. Proposed Method

In this section, we first describe how to generate high-quality bounding-box annotations for initial training samples (i.e., multitemporal RSIs). Then, the proposed DAL-based object detection method is detailed.

A. Initial Bounding-Box Annotation Generation

Change detection is the task of identifying differences in the state of an object or phenomenon by observing it at

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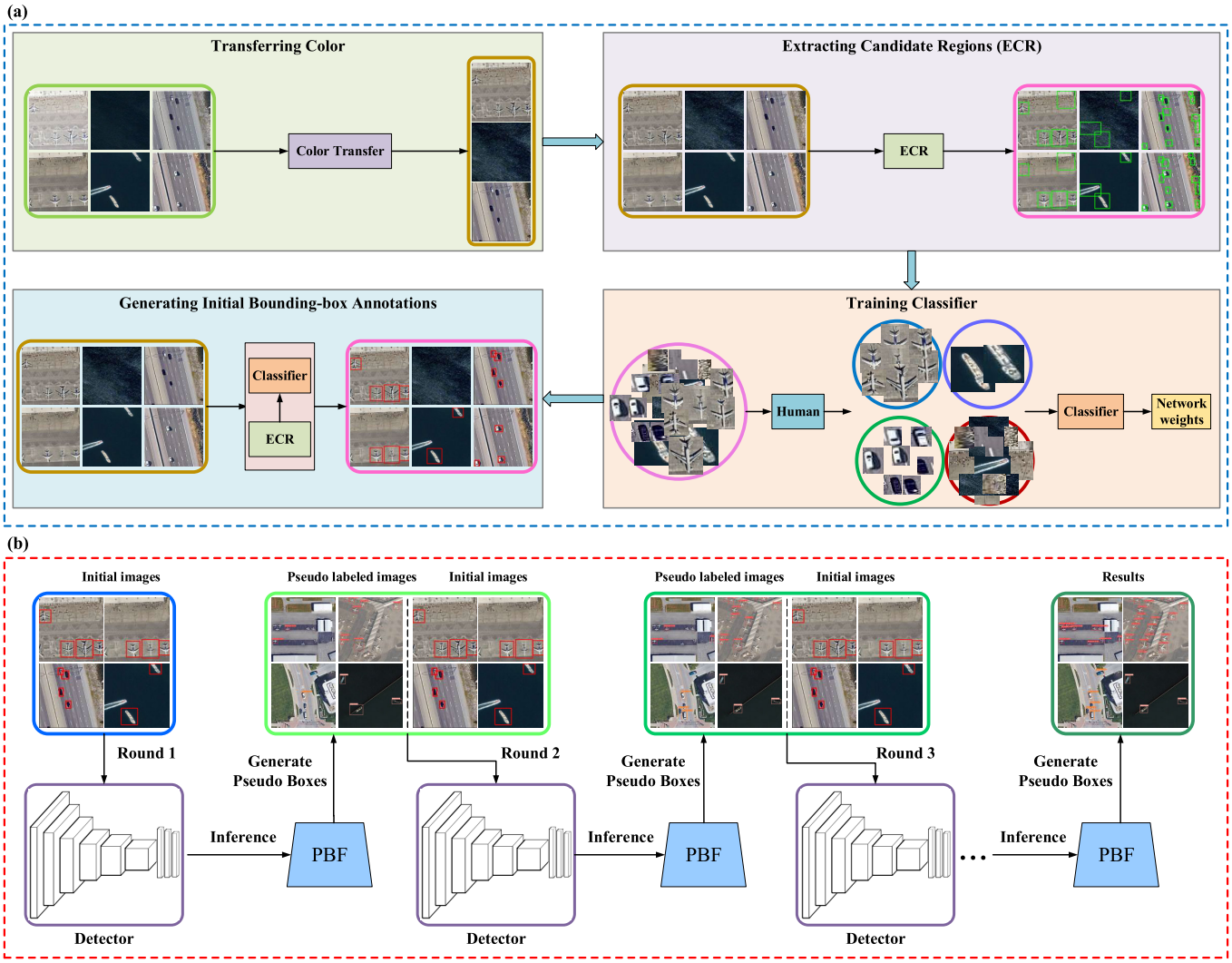


Fig. 2. Pipeline of the proposed DAL framework. PBF represents the proposed pseudobox filtering strategy, which is used to filter the unreliable labels of training instances. (a) Generating initial bounding-box annotations. (b) Producing pseudoboxes by dynamic automatic learning.

different times [37]. Inspired by this study, a novel strategy, named IAG, is developed in this article to generate high-quality bounding-box annotations for initial training samples (i.e., multitemporal RSIs containing movable objects). The framework and workflow of the IAG are outlined in Fig. 2 and Algorithm 1. The details of the relevant algorithms are described in the following.

1) Image Collection: In this article, we mainly study the detection of vehicle, airplane, and ship objects. We first obtain the longitude and latitude coordinates of the places (e.g., airports, harbors, highways, and park lots) where the target to be detected often appears on the Internet. For example, if you want to get the latitude and longitude coordinates of Shanghai Hongqiao Airport, you only need to search for "the latitude and longitude coordinates of Shanghai Hongqiao Airport" in the search engine, and relevant information (i.e., 121.33176 and 31.19459) will be fed back. Then, the BIGEMAP software is used to obtain multitemporal RSIs $N = \{(I_1t_1, I_1t_2), (I_2t_1, I_2t_2), \ldots, (I_nt_1, I_nt_2)\}$ [42] according to the collected longitude and latitude coordinates. In this way, the publicly available latitude and longitude coordinates can be used to avoid time consuming blindly looking for the target region and the historical image function of the BIGEMAP software can be used to save the cost of acquiring multitemporal images. It should be noted that we only use the BIGEMAP software to obtain multitemporal data, and the subsequent algorithms have nothing to do with the software. Researchers can also use ERDAS Imagine, ArcGIS Pro, and other tools that support English to obtain multitemporal data.

2) Image Preprocessing: To generate high-quality bounding-box annotations for initial training samples, the image preprocessing stage involves three key steps: color transfer, similarity evaluation, and image alignment.

a) Color transfer: The color difference of RSIs in different periods of the same area will seriously affect the accuracy of change detection. Inspired by the work of Reinhard et al. [38], we transfer the color characteristics of multitemporal RSIs to help detect changing regions. Specifically, for multitemporal RSIs $N = \{(I_1t_1, I_1t_2), (I_2t_1, I_2t_2), \ldots, (I_nt_1, I_nt_2)\}$, we consider I_it_1 as the source image and I_it_2 as the target image. For the source image I_it_1, the transferred color at pixel C_t in $l\alpha\beta$ color space

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Algorithm 1 Initial Bounding-Box Annotation Generation Strategy

Require: Multitemporal remote sensing images: $N = \{(I_1t_1, I_1t_2), (I_2t_1, I_2t_2), \ldots, (I_nt_1, I_nt_2)\}$.

Ensure: Bounding-box annotations.

1: Collect multitemporal remote sensing images (i.e., N) from the BIGEMAP software;
2: Use the color transfer method [38] to transfer color characteristics from a source image (e.g., I_it_1) to a target image (e.g., I_it_2);
3: Apply the structural similarity index measure (SSIM) index [39] as the criterion to evaluate the multitemporal images similarity to filter out images with large differences, and get new multitemporal images $M = \left\{(\widetilde{I_1t_1}, I_1t_2), (\widetilde{I_2t_1}, I_2t_2), \ldots, (\widetilde{I_mt_1}, I_mt_2)\right\}$;
4: Use the Enhanced Correlation Coefficient (ECC) [40] to achieve the image alignment and get new multitemporal remote sensing images $\overline{M} = \left\{(\widehat{I_1t_1}, I_1t_2), (\widehat{I_2t_1}, I_2t_2), \ldots, (\widehat{I_mt_1}, I_mt_2)\right\}$;
5: Execute the SSIM on $\overline{M}$ to detect changed objects from multitemporal remote sensing images and crop changed regions to generate the image set D;
6: Manually label images from D with image-level labels and adopt Super-Resolution Via Repeated Refinement (SR3) [41] algorithm to super-resolution these images, then use these super-resolution images with image-level labels to train a classifier C;
7: Execute the SSIM again on N and introduce the classifier C to obtain high-quality bounding-box annotations.

is expressed as follows:

$$g(C_t) = \mu_s + \frac{\sigma_s}{\sigma_t}(C_t - \mu_t) \tag{1}$$

where μ_s and μ_t represent the mean value of the Gaussian distribution of the source image and the target image in the $l\alpha\beta$ color space, respectively. σ_s, σ_t are the respective standard deviations [43]. Fig. 3 shows some examples of color transfer results.

b) Similarity evaluation: How to measure the similarity between images plays an important role in generating bounding-box annotations of initial training samples. If a pair of multitemporal RSIs differ greatly, too many nontarget areas will be generated, which may affect the quality and efficiency of generating bounding-box annotations. In this article, the structural similarity index measure (SSIM) index is employed as the criterion to evaluate the image similarity to filter out multitemporal RSIs with large differences. The SSIM index between two image patches $x = \{x_i | i = 1, \ldots, M\}$ and $y = \{y_i | i = 1, \ldots, M\}$ is computed as [44]

$$S(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{2}$$

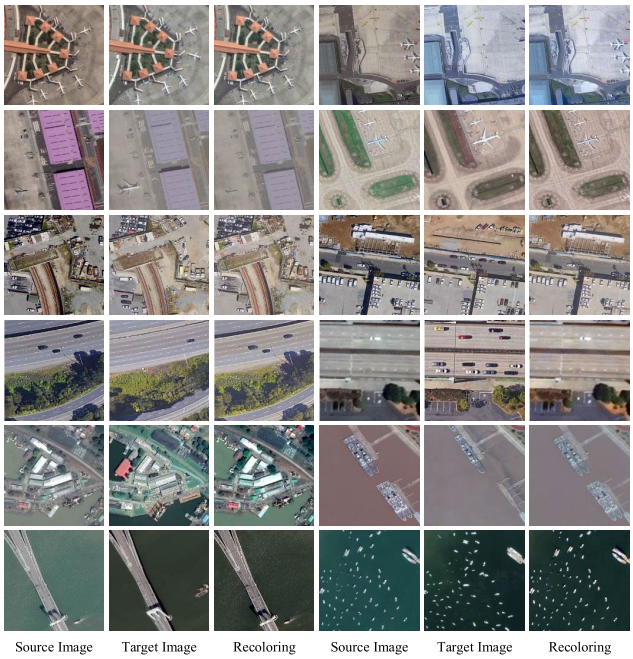


Fig. 3. Color transfer between images. Recoloring represents a new source image after applying the color transfer algorithm.

here

$$\mu_x = \frac{1}{M}\sum_{i=1}^{M} x_i, \quad \sigma_x^2 = \frac{1}{M}\sum_{i=1}^{M}(x_i - \mu_x)^2$$
$$\sigma_{xy} = \frac{1}{M}\sum_{i=1}^{M}(x_i - \mu_x)(y_i - \mu_y) \tag{3}$$

where C_1 and C_2 are positive constants. The maximum SSIM index is 1, which occurs when $x = y$ [39], [45]. In our experiments, the sliding window size of SSIM is 7×7, and image pairs with $S(x, y) > 0.3$ are used to participate in bounding-box annotations generation for initial training samples.

c) Image alignment: Unaligned multitemporal RSIs may be detrimental to change detection. In this article, the enhanced correlation coefficient (ECC) is used for image alignment.

3) Extracting Candidate Regions: In recent years, the SSIM index has also been widely used in the field of change detection [46]. The SSIM is more suitable for change detection using homogeneous images [47]. The multitemporal images we collected are homogeneous, and the background of these multitemporal data is relatively simple. To this end, we use the SSIM index to obtain candidate regions for initial training samples and adopt some strategies to ensure the quality of the obtained candidate regions. Specifically, we first use the SSIM index to obtain a large number of changed regions with irregular shapes. Then, we obtain regular regions by extracting the outer rectangle of the contour of irregular regions and use geometric constraints to filter out extremely unreliable changed regions. Next, the scale invariant feature transform (SIFT) operation [48] is performed on the changed regions of the same position of the two images, and the regions with many matching points are removed. Finally, the remaining

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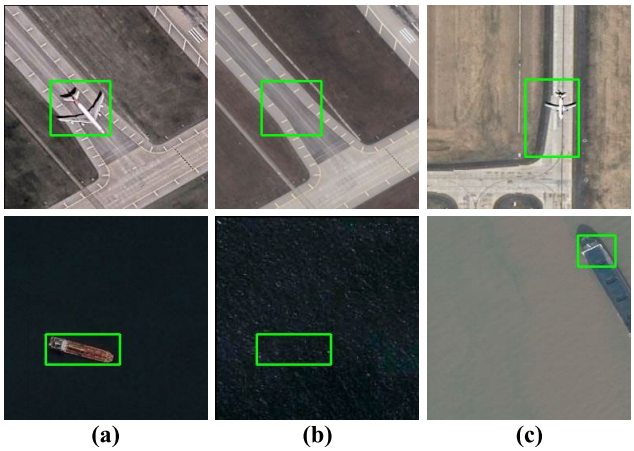


Fig. 4. Some examples of candidate region extraction results. (a) Unknown category. (b) Negative instances. (c) Positive instances of bad quality.

changed regions are selected as candidate regions. There are two aspects to note here: 1) objects usually only occupy a small proportion of the large-scale image with complex backgrounds in RSIs. Therefore, we use the area ratio S_{ratio} (i.e., the proportion of the area between the changed region and the whole image) as geometric constraints to filter out extremely unreliable changed regions. In our experiments, changed regions with S_{ratio} greater than 0.8 are filtered and 2) after performing the SIFT operation on the two changed regions, if there are many matching points in the two regions, indicating that two regions are relatively similar, then they can be considered as nonchanged regions. When the same object appears in two regions, there will be a large number of matching points, and we regard these two regions as nonchanging regions. Doing so can lead to missing true positive regions, whereas our method is an iterative learning process. In iterative training, the labels for the next round of training will be rereasoned on the unlabeled data by the model trained in the previous round to prevent the loss of true positive regions. In this article, we consider regions with matching points greater than 30 as nonchanged regions.

4) Generating Initial Bounding-Box Annotations: The bounding-box annotations of initial training samples cannot be directly obtained from the candidate region extraction process, mainly due to: 1) unknown category. The category label of the object in the candidate region cannot be known [see Fig. 4(a)]; 2) negative instances. There are no targets in some candidate regions [see Fig. 4(b)]; and 3) positive instances of bad quality. Some candidate regions are noisy, such as the object occupies a small part of the candidate region or the candidate region only occupies a part of the object [see Fig. 4(c)]. To this end, we use the following steps to obtain accurate bounding-box annotations of initial training samples. After obtaining candidate regions, we first cut candidate regions from the original images and manually labeled image-level labels for these images. In this article, we focus on three object categories (i.e., "airplane," "vehicle," and "ship") and one background category (i.e., "background") and directly label candidate regions without objects (i.e., vehicle, airplane, and ship) as the "background" category. Note that when collecting multitemporal data, including vehicles, airplanes, and ships, we collect them in groups by scene, and the candidate regions

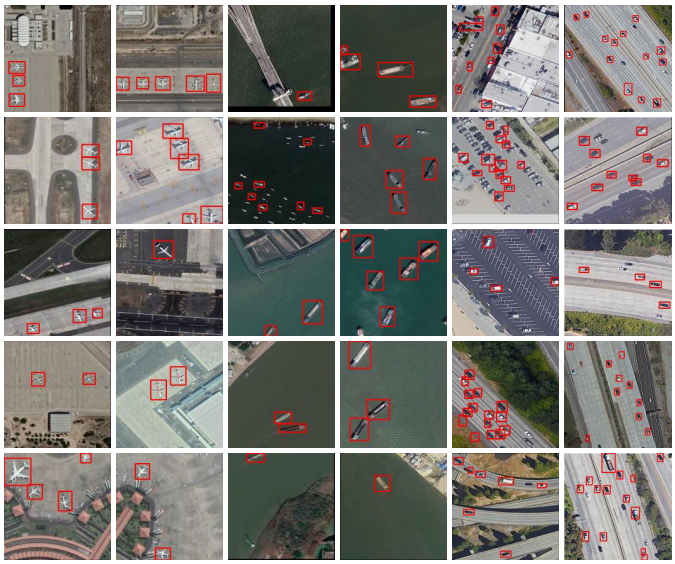


Fig. 5. Some examples of initial training samples with bounding-box annotations.

are also obtained in groups. Therefore, it is easy for us to label image-level labels for candidate images. For example, when the collected multitemporal data are a scene containing airplane objects, we only need to judge whether the region is "airplane" or "background" during image-level labeling for a candidate region. However, if the image contains multiple categories of objects, some true positive regions will be lost when the candidate regions are obtained by grouping. For example, when images containing airplane objects are collected from an airport scene, vehicles may also be present in these images, and we discard regions containing vehicles during initial bounding-box annotation generation. Our proposed DAL is an iterative learning method, which can rereason the unlabeled data during the iterative learning process to prevent the loss of true positive regions. These images are then used to train a fully supervised classifier. Next, changed regions are obtained by III-A3, and positive instances of bad quality and negative instances are filtered by the trained classifier. In other words, bad quality and negative instances are removed by the classifier. Finally, the bounding-box annotations corresponding to the remaining changed regions can be used as the annotations for the initial training samples. Fig. 5 shows some samples from the initial training images with bounding-box annotations.

Objects in RSIs are usually small, which is not conducive to the classifier to fully mine the discriminative information of objects, so it is difficult for the classifier to effectively filter out the noisy candidate regions. Thus, we use the SR3 algorithm to super-resolution candidate regions to improve the performance of the classifier. More specifically, we use the SR3 algorithm to super-resolution candidate regions and then use the super-resolution image to train the classifier (i.e., ResNet-50 [49]), thereby filtering out noisy candidate regions whose predicted label (i.e., image-level label) is "background." In this article, we refer to this way of filtering noisy regions as PBF. It should be emphasized that the learned classifier can filter out noisy pseudoboxes without retraining in the subsequent iterative learning process. Although the proposed

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Algorithm 2 DAL for Object Detection

Require: Initial training samples with bounding-box annotations $D^l = \{(x_i, y_i)\}_{i=1}^{P}$, unlabeled dataset $D^u = \{u_i\}_{i=1}^{Q}$.

Ensure: Learned detection model.

1: Train a teacher model $f(x, \theta_*^t)$ on initial training samples D^l;
2: Generate pseudoboxes on unlabeled images D^u with $f(u, \theta_*^t)$ to obtain $\widetilde{D}^l = \{(u_i, f(u_i, \theta_*^t))\}_{i=1}^{Q}$;
3: Apply the PBF strategy to filter out unreliable pseudoboxes to obtain $\widehat{D}^l = \{(u_i, f(u_i, \theta_*^t))\}_{i=1}^{K}$, where $K \leq Q$;
4: Retrain a student model on the union set $D^l \cup \widehat{D}^l$;
5: Iterative training: Use the student as a teacher and go back to step 2 until the number of instances of training images no longer grows.

IAG needs to set some parameters and label image-level labels for some images, its cost is negligible compared with labeling bounding-box annotations for a large amount of training data.

B. Dynamic Automatic Learning

The framework and workflow of the proposed DAL are outlined in Fig. 2 and Algorithm 2. Given a set of labeled images (i.e., initial training samples with bounding-box annotations) $D^l = \{(x_i, y_i)\}_{i=1}^{P}$ and a set of unlabeled images $D^u = \{u_i\}_{i=1}^{Q}$, where y_i represents the labeled bounding boxes. Following some previous works [33], [50], [51], [52], a teacher object detector is first trained on the initial training samples with bounding-box annotations D^l. Then, we use the trained teacher model $f(x, \theta_*^t)$ to produce pseudoboxes for each unlabeled image $u_i \in D^u$ to obtain $\widetilde{D}^l = \{(u_i, f(u_i, \theta_*^t))\}_{i=1}^{Q}$. Next, these pseudoboxes are further filtered by our proposed PBF, and the remaining pseudoboxes and corresponding images constitute a new training subset $\widehat{D}^l$. After that, learn a new student model on the union set $D^l \cup \widehat{D}^l$. The teacher generates pseudoboxes to train the student, and the student updates what he/she has learned back to the teacher, iterating this process until the number of instances of the training images no longer grows. By continuously learning from the training samples with bounding-box annotations generated by the previous iteration, the detection network is gradually optimized, and the generated pseudoboxes will become more accurate gradually. Following this process, we cannot only reject the noisy instances but also maintain the diversity of the training samples.

C. Pseudobox Filtering

Our proposed DAL approach is heavily affected by the quality of pseudoboxes. Once the noisy pseudoboxes are selected, incorrect predictions will be amplified in iterative learning. Some self-training approaches [50], [53] usually set thresholds to filter out predictions with low classification confidence. However, several studies have shown that the correlation between the quality of pseudoboxes and their confidence scores is not high [33], [52], [54]. As a solution, we use our proposed PBF to remove the influence of positive instances of bad quality and negative instances during iterative learning. Specifically, we use the SR3 algorithm to super-resolution each pseudobox image during each round of reasoning, and then use the learned classifier to predict the pseudobox images. Finally, we remove pseudoboxes where the pseudolabels (i.e., image-level labels) produced by the teacher model are inconsistent with the labels predicted by the classifier. Following this process, we cannot only reject pseudoboxes that are incorrectly classified but also filter out pseudoboxes that are inaccurately localized. Besides, in iterative training, the labels for the next round of training will be rereasoned on the unlabeled data by the model trained in the previous round to prevent the loss of true positive regions.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Metric

1) Evaluation Datasets: We validated the effectiveness of our method on two challenging datasets: NWPU VHR-10.v2 and DIOR datasets.

a) NWPU VHR-10.v2: The NWPU VHR-10.v2 dataset contains 650 images with different sizes (i.e., from 533 × 597 to 1728 × 1028 pixels), of which 231 images contain vehicle, airplane, and ship objects. According to the dataset division criteria of some previous works [11], [12], [15], we randomly selected 56 images containing vehicle, airplane, and ship objects as test data and the remaining 594 images as unlabeled data.

b) DIOR: The DIOR dataset contains 23 463 images of 20 object categories, which are divided into training, test, and validation sets. In this article, we use the training and validation sets of DIOR as unlabeled data (i.e., 11 725 images) and use images in the test set of the DIOR containing vehicle, airplane, and ship objects (i.e., 4706 images) to verify the performance of our proposed method.

2) Multitemporal Remote Sensing Images: The NWU-ABC-MRSI dataset contains 10 242 multitemporal RSIs of 3 object categories (i.e., vehicle, airplane, and ship). After the initial bounding-box annotation generation process, the remaining 4231 images with bounding-box annotations are regarded as the initial training samples. The NWU-ABC-MRSI dataset is publicly shared at https://github.com/XZNWU/NWU-ABC-MRSI/.

3) Unlabeled Datasets: We obtain unlabeled images from the RSOD [55], LEarning, VIsion and Remote sensing laboratory (LEVIR) [56], DOTA-v1.0 training set [1], High Resolution Ship Collection 2016 (HRSC2016) [57], NWPU VHR-10.v2 trainval set [17], DIOR trainval set [4], NWU-ABC-MRSI, and University of Chinese Academy of Sciences-Aerial Object Detection (UCAS-AOD) [58]. Although the images in these datasets are labeled, we regard them as unlabeled data. Table I shows the size of the unlabeled datasets. It should be noted that in each round of inference, all unlabeled data will be used simultaneously to generate pseudoboxes for the next round of training data.

a) RSOD: The RSOD dataset contains 976 images of four object categories (i.e., airplane, oil tank, playground, and overpass). In this article, all images in this dataset are treated as unlabeled data.

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TABLE I
UNLABELED DATASETS IN THE EXPERIMENTS

Datasets	NWPU VHR-10.v2	DIOR	NWU-ABC-MRSI	RSOD	LEVIR	HRSC2016	UCAS-AOD	DOTA-v1.0
Number	594	11,725	10,242	976	22,000	617	1,510	36,816

TABLE II
AP (%) PERFORMANCE ON DIOR AND NWPU VHR-10.v2 TEST SETS, RESPECTIVELY. NOTE THAT THE NUMBER OF INSTANCES IN THE TABLE IS APPROXIMATE

Methods	Iteration	Train		Val		DIOR				NWPU VHR-10.v2			
		images	instances	images	instances	Aeroplane	Ship	Vehicle	mAP	Aeroplane	Ship	Vehicle	mAP
w/o SR3	iter1	3,224	7,210	1,030	2,300	0.155	0.0103	0.0265	0.0641	0.211	0.135	0.045	0.130
w/ SR3	iter1	3,211	6,720	1,020	2,200	0.162	0.0112	0.0303	0.0679	0.319	0.225	0.015	0.186

b) LEVIR: The LEVIR dataset contains 22 000 images covering three object categories (i.e., airplane, ship, and storage tank). In our experiments, all images in this dataset are treated as unlabeled data.

c) HRSC2016: The HRSC2016 is a ship detection dataset. It has 1061 images, of which 436 are for training, 181 for validation, and 444 for testing. In this article, we use the training and validation sets of HRSC2016 as unlabeled data.

d) UCAS-AOD: In UCAS-AOD, there are two categories: airplane and vehicle. It consists of 1000 airplane images containing 7482 instances and 510 vehicle images containing 7114 instances. In our experiments, all images in this dataset are treated as unlabeled data.

e) DOTA-v1.0: The DOTA-v1.0 contains 2806 images with size ranges from 800 × 800 to 4000 × 4000. For accurate detection, the images are cropped into 640 × 640 patches with a stride of 100. The training and validation sets contain 36 816 and 12 493 images after the cropping, respectively. In our experiments, we use the training set of DOTA-v1.0 as unlabeled data.

4) Evaluation Metric: The average precision (AP) is used to measure the performance of object detection. This metric treats each detection result as a positive detection when the intersection over union (IoU) is greater than 0.5 [59].

B. Implementation Details

The model of You Only Look Once v5x (YOLOv5x) [60] for object detection can achieve a good tradeoff between accuracy and speed. In this article, we choose YOLOv5x as the object detector. During training, the batch size is set to 8 with an initial learning rate of 0.001, weight decay of 0.0005, and momentum of 0.9. The training step is set to 100k steps, and the learning rate will be decreased by a factor of 10 after every 20K steps. In our experiments, we use ResNet50 as our classifier to perform noisy PBF. We trained the ResNet50 using stochastic gradient descent with a batch size of 8, a weight decay of 0.0005, and a momentum of 0.9. The learning rates of the backbone layers are set to 0.0002 and are reduced by following the cosine annealing schedule [61]. When training the classifier, the input images are adjusted to a fixed size of 128 × 128. The proposed DAL is implemented using PyTorch, which is powered on an NVIDIA RTX 2080 Ti GPU.

C. Importance of Filtering Noisy Pseudoboxes

We investigate the impact of image super-resolution on performance during the initial bounding-box annotation generation stage. To validate the effectiveness of the image super-resolution, we removed the SR3 from our PBF. Specifically, after removing SR3 in PBF, 4254 images in the NWU-ABC-MRSI dataset are used for experiments. In the case of using SR3, 4231 images in the NWU-ABC-MRSI dataset are used for experiments. In Table II, we provide the experimental results with (w/ SR3) and without SR3 (w/o SR3), where the training set is NWU-ABC-MRSI, and the test data comes from the test sets of DIOR and NWPU VHR-10.v2, respectively. One can easily observe that image super-resolution plays an important role in filtering out noisy pseudoboxes (6.41% versus 6.79% and 13.0% versus 18.6%). One possible reason is that the objects in RSIs are usually small, which is not conducive for the classifier to fully mine the discriminative information of the objects, resulting in the difficulty for the classifier to effectively filter noisy pseudoboxes. Note that the number of test images of the detection model using SR3 and not using SR3 is the same, and they all use the test set of DIOR and NWPU VHR-10.v2 to verify the performance of the algorithm. However, since some images with imprecise labels will be filtered out after using SR3, the number of images in the training and validation sets of the two models is different.

The PBF strategy aims to purify the quality of pseudoboxes by removing the influence of positive instances of bad quality and negative instances. To verify the effectiveness of the proposed PBF strategy during iterative learning, we compare the experimental results with (w/PBF) and without (w/o PBF) PBF. Table III shows the evaluation results on the NWPU VHR-10.v2 and DIOR datasets. One can easily observe that w/PBF can offer 3.3% mAP gains on the DIOR dataset (33.2% versus 36.5%) and 3.3% mAP gains on the NWPU VHR-10.v2 dataset (60.6% versus 63.9%) compared with w/o PBF.

Traditional self-training approaches [50], [53] usually use a confidence score to remove noise predictions to select a subset of prediction labels of unlabeled data for model training. It is a very interesting experiment to compare our proposed PBF with the method of filtering noisy pseudoboxes based on confidence scores. Following some previous works [50], [53], we set the confidence score to 0.7, that is, only pseudoboxes with a confidence score higher than 0.7 are retained. As shown in Table IV, the accuracy of using PBF (w/ PBF) is higher

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TABLE III

AP (%) PERFORMANCE ON DIOR AND NWPU VHR-10.v2 TEST SET. THE FIRST ROUND OF TRAINING FOR ALL METHODS DIRECTLY USES INITIAL TRAINING IMAGES (I.E., 4231 IMAGES WITH BOUNDING-BOX ANNOTATIONS) AS THE TRAINING DATA

Methods	Iteration	Train		Val		DIOR				NWPU VHR-10.v2			
		images	instances	images	instances	Aeroplane	Ship	Vehicle	mAP	Aeroplane	Ship	Vehicle	mAP
w/o PBF	iter1	3,211	6,720	1,020	2,200	0.162	0.0112	0.0303	0.0679	0.319	0.225	0.015	0.186
	iter2	19,345	37,000	1,869	3,360	0.272	0.0176	0.0504	0.113	0.403	0.341	0.0487	0.264
	iter3	28,234	60,500	3,073	6,170	0.375	0.0240	0.0760	0.158	0.503	0.510	0.128	0.380
	iter4	37,017	92,100	4,466	10,400	0.479	0.0300	0.0901	0.200	0.622	0.594	0.146	0.454
	iter5	43,320	121,000	5,670	15,300	0.612	0.0357	0.1080	0.252	0.706	0.676	0.166	0.516
	iter6	44,645	143,000	5,812	17,200	0.686	0.0354	0.1230	0.281	0.808	0.707	0.144	0.553
	iter7	46,059	160,000	6,128	20,100	0.757	0.0408	0.1380	0.312	**0.896**	**0.751**	**0.172**	**0.606**
	iter8	47,908	178,000	6,534	22,600	0.810	0.0450	0.1320	0.329	0.952	0.728	0.124	0.601
	iter9	51,784	223,000	7,549	30,100	0.816	0.0419	0.1280	0.329	0.947	0.677	0.143	0.589
	iter10	48,644	206,000	6,884	28,100	**0.826**	**0.0420**	**0.1280**	**0.332**	0.948	0.676	0.146	0.590
w/ PBF	iter1	3,211	6,720	1,020	2,200	0.162	0.0112	0.0303	0.0679	0.319	0.225	0.015	0.186
	iter2	17,755	31,700	1,549	2,540	0.292	0.0190	0.0463	0.119	0.437	0.430	0.089	0.319
	iter3	23,204	46,400	2,266	4,190	0.395	0.0261	0.0693	0.164	0.504	0.523	0.165	0.397
	iter4	28,699	63,400	3,077	6,600	0.521	0.0333	0.0911	0.215	0.632	0.581	0.259	0.491
	iter5	32,469	81,700	3,638	9,010	0.632	0.0405	0.0898	0.254	0.804	0.560	0.227	0.533
	iter6	31,294	83,000	3,553	9,280	0.744	0.0553	0.1080	0.303	0.869	0.580	0.309	0.586
	iter7	34,125	100,000	3,988	11,900	0.820	0.0928	0.1670	0.336	0.887	0.587	0.342	0.605
	iter8	36,376	121,000	4,343	14,900	0.828	0.0901	0.1510	0.357	**0.920**	**0.580**	**0.418**	**0.639**
	iter9	37,765	137,000	4,569	17,700	**0.823**	**0.1150**	**0.1560**	**0.365**	0.935	0.570	0.386	0.630
	iter10	35,965	144,000	4,186	144,00	0.817	0.0891	0.1510	0.352	0.844	0.422	0.217	0.494

TABLE IV

AP (%) PERFORMANCE ON DIOR AND NWPU VHR-10.v2 TEST SET. THE FIRST ROUND OF TRAINING FOR ALL METHODS DIRECTLY USES INITIAL TRAINING IMAGES AS THE TRAINING DATA

Methods	Iteration	DIOR				NWPU VHR-10.v2			
		Aeroplane	Ship	Vehicle	mAP	Aeroplane	Ship	Vehicle	mAP
$score \geq 0.7$	iter1	0.162	0.0112	0.0303	0.0679	0.319	0.225	0.015	0.186
	iter2	0.223	0.018	0.0484	0.0965	0.345	0.369	0.075	0.263
	iter3	0.266	0.0218	0.0699	0.119	0.407	0.425	0.114	0.315
w/ PBF	iter1	0.162	0.0112	0.0303	0.0679	0.319	0.225	0.015	0.186
	iter2	0.292	0.0190	0.0463	0.119	0.437	0.430	0.089	0.319
	iter3	0.395	0.0261	0.0693	0.164	0.504	0.523	0.165	0.397

TABLE V

PERFORMANCE OF OUR PROPOSED METHOD UNDER DIFFERENT INITIAL TRAINING DATASET SIZES

Iteration	20% dataset		50% dataset		100% dataset	
	DIOR	NWPU-VHR-10.v2	DIOR	NWPU-VHR-10.v2	DIOR	NWPU-VHR-10.v2
iter1	0.049	0.122	0.066	0.122	0.068	0.186
iter2	0.098	0.257	0.122	0.266	0.119	0.319
iter3	0.113	0.314	0.152	0.373	0.164	0.397
iter4	0.135	0.380	0.204	0.472	0.215	0.491
iter5	0.146	0.418	0.235	0.516	0.254	0.533
iter6	0.171	0.457	0.293	0.563	0.303	0.586
iter7	0.189	0.498	0.324	0.581	0.336	0.605
iter8	0.194	0.489	0.338	0.615	0.357	0.639
iter9	0.200	0.513	0.346	0.604	0.365	0.630
iter10	0.210	0.521	0.323	0.593	0.352	0.494

than using only pseudoboxes whose confidence scores are higher than 0.7. The reason is that the correlation between the quality of pseudoboxes and their confidence scores is not high [33], [52], [54]. To be specific, poor predictions with high confidence scores as training data will seriously affect the performance of the model, and good predictions with low confidence scores are removed before the training detector will result in the underutilization of unlabeled data.

D. Importance of Initial Training Sample Size

In this section, we analyze the impact of the number of initial training samples on the performance of detectors in iterative learning. In Table V, we compare the experimental results using all initial training samples (100% dataset), 50% initial training samples (50% dataset), and 20% initial training samples (20% dataset). We can see that the performance of the detector can be improved when more and more initial training samples are used for network training.

E. Select the Number of Iterations

There is no mechanism to guarantee the convergence of the iteration. The iterative procedure simply stopped when the number of instances of training images stops growing. In addition, except for the 1020 images from the NWU-ABC-MRSI dataset in the first round, the subsequent validation sets are all from the DOTA-v1.0 validation set, and the bounding-box annotations of these images are all inferred from

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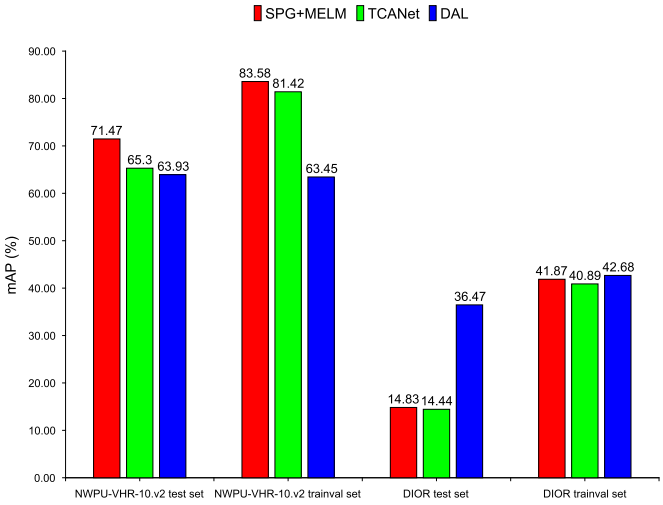


Fig. 6. Performance comparisons of different methods in terms of AP values.

TABLE VI

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE NWPU-VHR-10.V2 TEST SET

Methods	Aeroplane	Ship	Vehicle	mAP
Fast-RCNN [62]	0.9091	0.9060	0.6984	0.8378
Faster-RCNN [63]	0.9090	0.8630	0.7814	0.8511
RICNN [64]	0.8871	0.7834	0.7201	0.7969
RCNN [65]	0.8537	0.8888	0.4992	0.7472
Transfer CNN [66]	0.6603	0.5713	0.4127	0.5481
COPD [67]	0.6225	0.6937	0.4428	0.5863
RICO [17]	0.9970	0.9080	0.8714	0.9255
YOLOv5x [60]	0.9940	0.8200	0.6100	0.8080
WSDDN [68]	0.3008	0.4172	0.0360	0.2513
OICR [69]	0.1366	0.6735	0.0373	0.2825
PCL [9]	0.2600	0.6376	0.1567	0.3514
MELM [70]	0.8086	0.6930	0.0868	0.5295
DCL [12]	0.7270	0.7425	0.3500	0.6065
TCANet [11]	0.8943	0.7818	0.2830	0.6530
DPLG [13]	0.8090	0.7830	0.0870	0.5597
MIG [14]	0.8869	0.7161	0.0906	0.5645
OS [15]	0.8164	0.6833	0.2587	0.5861
SPG+OICR [36]	0.4280	0.7256	0.2475	0.4670
SPG+MELM [36]	0.9042	**0.8100**	**0.4299**	**0.7147**
Ours	**0.9200**	0.5800	0.4180	0.6393

the previous round. Table III shows that as the number of iterations increases, the performance of all networks accordingly. In particular, performances rapidly increase until nine iterations, after which they slightly decrease. The reasons are twofold: 1) noisy pseudoboxes can be reduced but not entirely eliminated by our PBF strategy; 2) after several iterations, the student detector did not receive more useful supervision than the teacher provided. Fig. 7 shows some detection results of each iteration of our proposed approach. As can be seen, models can gradually improve their performance to localize more accurate instances.

F. Comparison With Weakly Supervised Methods

We compare the performance of the proposed method with WSOD methods designed for RSIs. For each object class, we provide experimental results on the NWPU VHR-10.v2 and DIOR datasets in Tables VI–IX. Note that the results of the comparison methods except the YOLOv5x are from the original papers. Experimental results show that our method achieves competitive results compared with state-of-the-art methods. The reasons may come from the following aspects: 1) existing WSOD approaches lack instance-level bounding-box supervision, which leads to the ambiguity of learning discriminative object detection models. However, our proposed IAG can effortlessly generate bounding-box annotations of initial training samples; 2) existing WSOD approaches are prone to suffer from model drift problems when unreliable labels are introduced in the iterative learning process. Our proposed PBF strategy can purify the quality of pseudoboxes to alleviate the model drift problem during the training process; and 3) our proposed DAL can mine valuable information from large-scale unlabeled data to enhance the learning of detection knowledge. Interestingly, we observe that our proposed DAL

TABLE VII

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE NWPU-VHR-10.V2 TRAINVAL SET

Methods	Aeroplane	Ship	Vehicle	mAP
WSDDN [68]	0.2232	0.3681	0.0289	0.2067
OICR [69]	0.2941	0.8333	0.1444	0.4239
PCL [9]	0.1176	0.5000	0.1556	0.2577
MELM [70]	0.8596	0.7742	0.1463	0.5934
TCANet [11]	0.9691	0.9178	0.5556	0.8142
DPLG [13]	0.8720	0.8510	0.1630	0.6287
MIG [14]	0.9779	0.9026	0.2157	0.6987
SPG+OICR [36]	0.4473	0.8829	0.2700	0.5334
SPG+MELM [36]	**0.9806**	**0.9267**	**0.6002**	**0.8358**
Ours	0.9325	0.6480	0.3225	0.6345

TABLE VIII

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE DIOR TEST SET

Methods	Aeroplane	Ship	Vehicle	mAP
Fast-RCNN [62]	0.4417	0.3213	0.1920	0.3183
Faster-RCNN [63]	0.5028	0.5848	0.3477	0.4784
YOLOv5x [60]	0.6890	0.7490	0.3950	0.6110
WSDDN [68]	0.0906	0.0070	0.0455	0.0477
OICR [69]	0.0870	0.0909	0.0909	0.0896
PCL [9]	0.2152	0.0909	0.0455	0.1172
MELM [70]	0.2814	0.0909	0.0004	0.1242
DCL [12]	0.2089	0.0909	0.0727	0.1242
TCANet [11]	0.2513	0.0909	0.0909	0.1444
MIG [14]	0.2220	0.0642	0.0023	0.0962
OS [15]	0.4110	**0.2134**	**0.1834**	0.2693
SPG+OICR [36]	0.1280	0.0923	0.0912	0.1038
SPG+MELM [36]	0.3132	0.1002	0.0315	0.1483
Ours	**0.8230**	0.1150	0.1560	**0.3647**

TABLE IX

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE DIOR TRAINVAL SET

Methods	Aeroplane	Ship	Vehicle	mAP
WSDDN [68]	0.0572	0.2296	0.1156	0.1341
OICR [69]	0.1598	0.3180	0.1583	0.2120
PCL [9]	0.6114	**0.3794**	0.1050	0.3653
MELM [70]	0.7698	0.2819	0.0462	0.3660
TCANet [11]	0.8158	0.3063	0.1045	0.4089
MIG [14]	0.7698	**0.3794**	0.0105	0.3866
SPG+OICR [36]	0.3506	0.3365	0.1622	0.2831
SPG+MELM [36]	0.8048	0.3516	0.0998	0.4187
Ours	**0.8970**	0.1280	**0.2555**	**0.4268**

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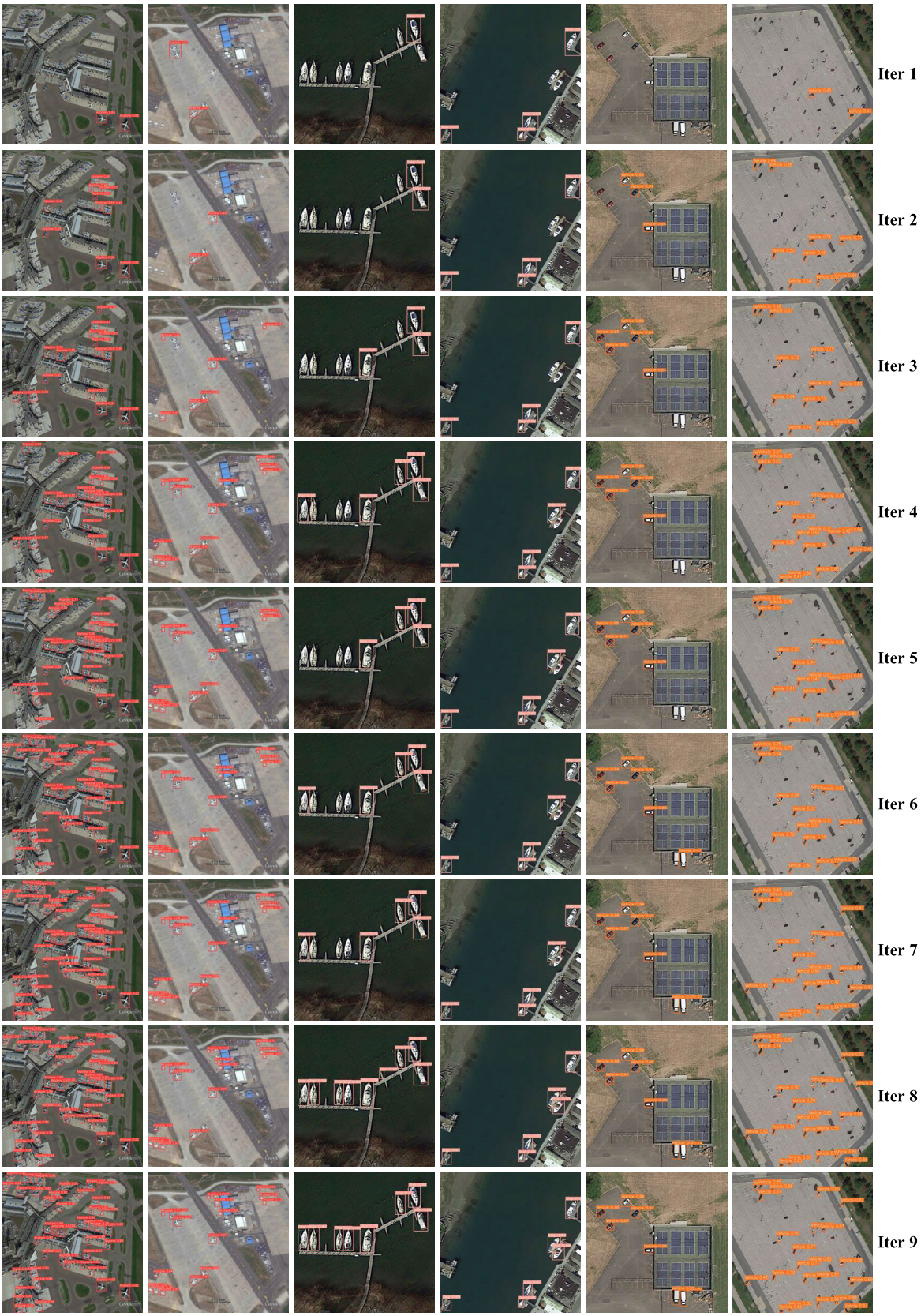


Fig. 7. Object detection results as iterations increase.

outperforms all previous methods on the DIOR dataset, while it performs slightly worse than TCANet and SPG + MELM on the NWPU VHR-10.v2 dataset (see Fig. 6). We think the reason is that images of the NWPU VHR-10.v2 dataset have much simpler and clearer backgrounds than DIOR [12], [36]. More specifically, the proposed DAL augments the training

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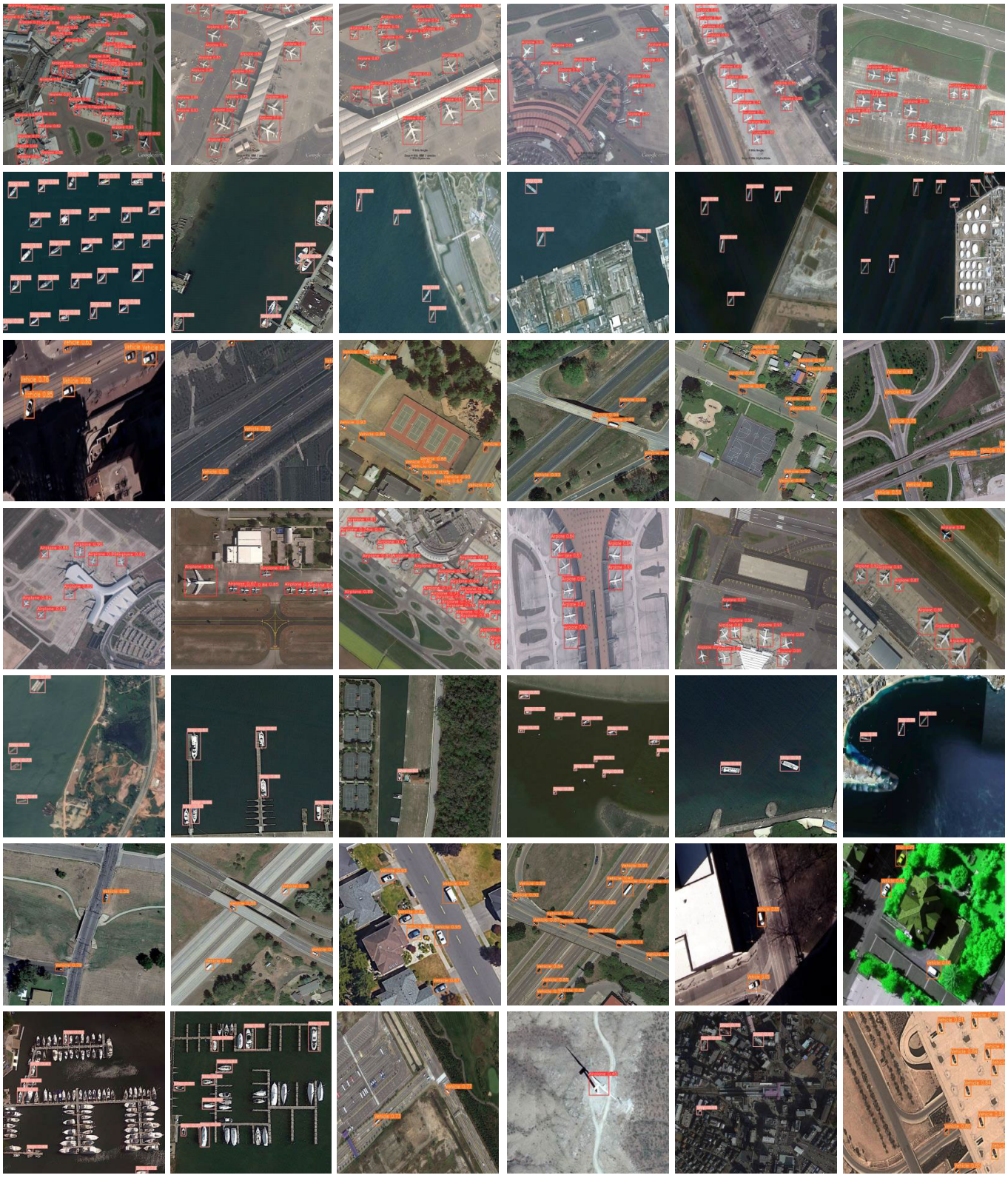


Fig. 8. Some detection results. The top six rows are success cases. The last row indicates some failure cases.

images by introducing a large amount of unlabeled data, thereby enhancing the generalization of the model. However, locating object in images with simple backgrounds is relatively easier and thus might not obviously benefit from models with strong generalization. Besides, in contrast to TCANet and SPG + MELM, our method does not use any labels from the target data (i.e., NWPU VHR-10.v2 and DIOR datasets), which is more challenging than image-level WSOD. From

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Tables III, VI, and VIII, we can see that our proposed method can achieve better results even without using PBF and using PBF can further improve the detection performance.

G. Comparison With Fully Supervised Methods

We further compare our DAL with eight fully supervised object detection approaches, including fast-RCNN [62], faster-RCNN [63], rotation-invariant CNN (RICNN) [64], RCNN [65], transfer CNN [66], collection of part detectors (COPD) [67], rotation-insensitive and context-augmented object detection (RICO) [17], and YOLOv5x. As we can capture in Tables VI and VIII, our proposed method can narrow the supervision gap between fully and WSOD. Especially, for the object class of airplanes, our method can achieve comparable or even superior performance than most fully supervised detectors. The main reason is that manually labeling large-scale images at the bounding-box level is expensive, resulting in a limited amount of data available for these fully supervised approaches. However, our proposed method can augment the training data to improve the performance of the model by using a large number of unlabeled images that are readily available.

H. Qualitative Results

As illustrated in Fig. 8, the top three rows show the detection results of our method on unlabeled images, and rows four through six show the detection results of our method on the NWPU VHR-10.v2 and DIOR datasets test sets. As can be observed, our method can accurately detect and locate each instance in the images. However, due to the lack of labels (i.e., image-level labels or bounding-box annotations) of the target data, the proposed method remains a challenge to address the problem of dense objects and the coexisting scene between objects and background (see the last row in Fig. 8). In the future, we plan to alleviate the aforementioned challenges.

V. Conclusion

In this article, a novel DAL approach is proposed to perform object detection for RSIs, which can progressively learn detection knowledge by alternating between detector improvement and reliable pseudobox generation. To obtain bounding-box annotations of initial training samples, we propose a novel IAG strategy, which can generate bounding-box annotations by leveraging the change detection technique. To remove the influence of positive instances of bad quality and negative instances, an effective PBF strategy is developed. Extensive experiments demonstrate that our proposed method can significantly improve the accuracy of object detection compared with state-of-the-art methods.

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